Name: ______________________________

Class: ______________________________

Date: ______________________________

Mr. Rawson • GCSE Physics

Homework — Series and Parallel Circuits

Every question here can be answered with $V = IR$ and the rules you measured on Monday. Show every step: an answer with no working earns almost nothing in the exam, and I cannot see where you went wrong.

1) State the two rules for a series circuit — one about current, one about potential difference — and the equation for the total resistance of two resistors in series. **[3]**

2) Calculate the total resistance, and then the current, in a series circuit where a $4\,\Omega$ resistor and a $2\,\Omega$ resistor are connected to a $12\,\mathrm{V}$ supply. **[3]**

3) Calculate the potential difference across *each* resistor in the circuit from question 2, and show that your two answers are consistent with the supply. **[3]**

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4) **Calculate** the current in each branch, and the total current from the supply, when a $3\,\Omega$ resistor and a $6\,\Omega$ resistor are connected *in parallel* across a $6\,\mathrm{V}$ supply. **[4]**

Parallel: Each branch gets the *full* supply pd — so use $V = IR$ on one branch at a time. You are *not* being asked for a total resistance: Combined Science never asks you to calculate that for parallel resistors.

5) **Explain** why adding a resistor *in series* increases the total resistance of a circuit, but adding a resistor *in parallel* decreases it. No calculation is needed. **[4]**

Think: How many routes can the charge take in each case?

6) **Explain** why the lights in a house are wired in parallel rather than in series. Refer to potential difference in your answer. **[4]**

Total: 21 marks. Due Tuesday 8 September — bring it to the lesson. If you are stuck, start from $V = IR$ and the rules table on Monday's sheet.